Kernix 0.11.1 / Performance brief

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

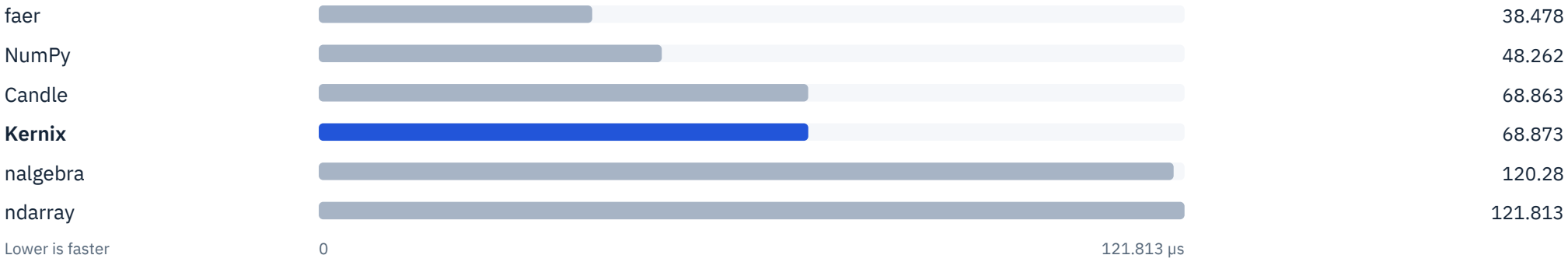
46 workloads

149 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.1 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

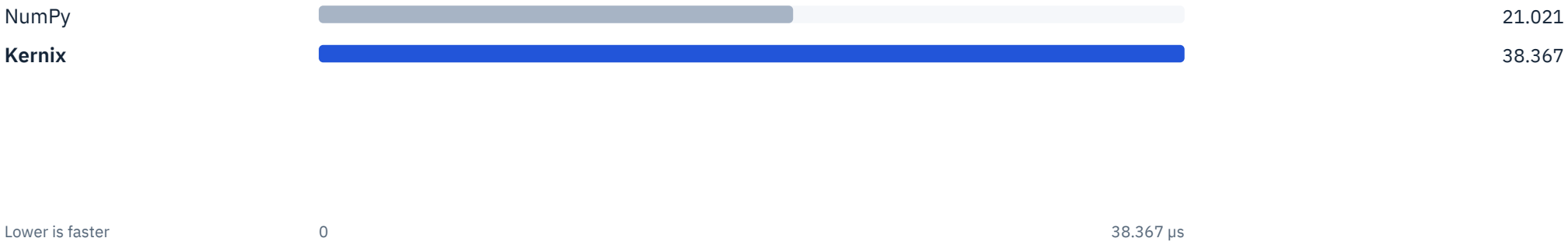
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.11.1 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

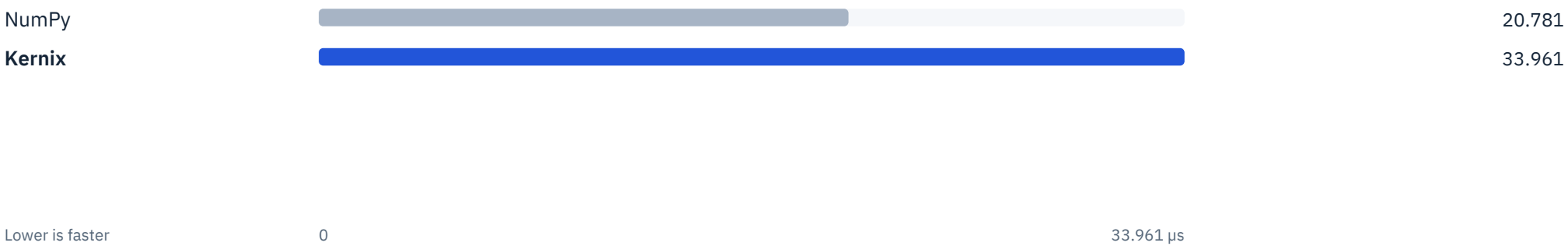
128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



Kernix 0.11.1 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

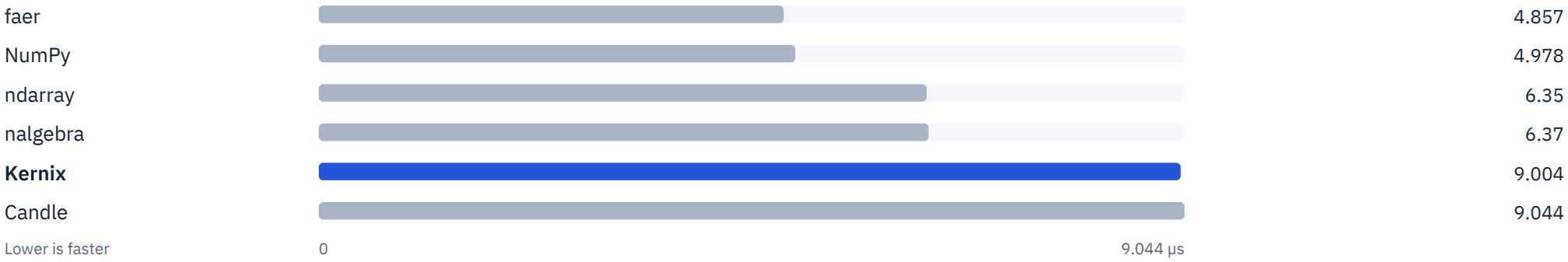
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



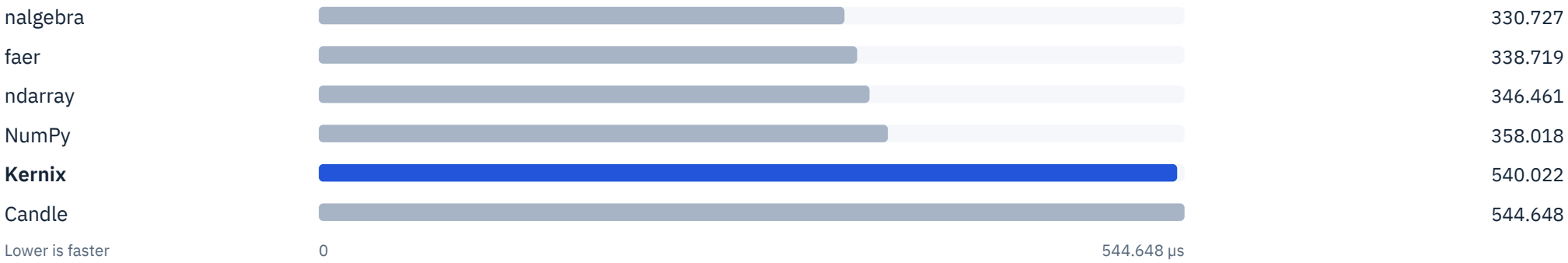
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



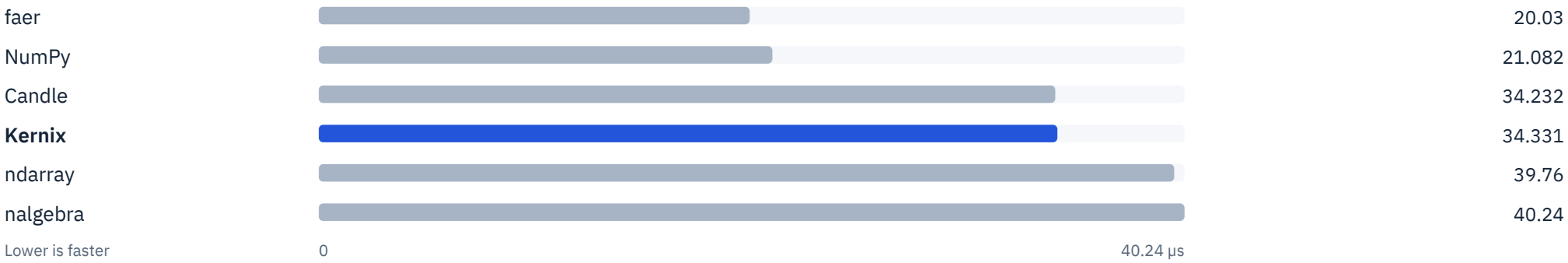
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

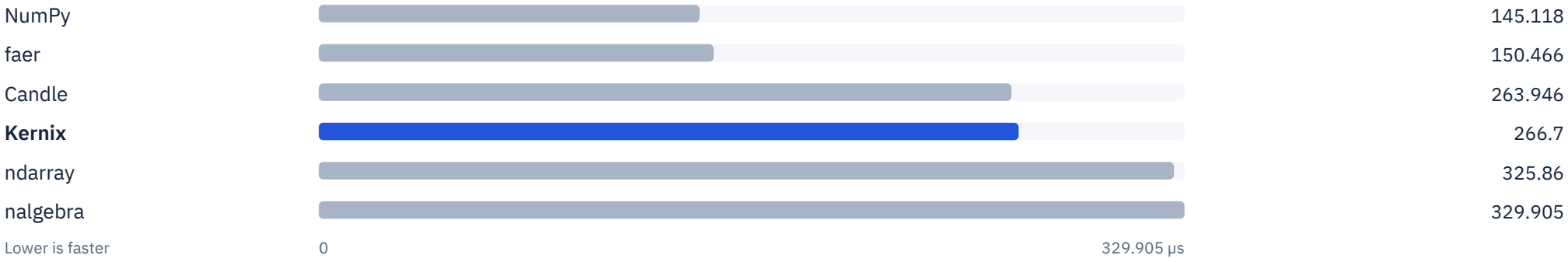
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.11.1 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

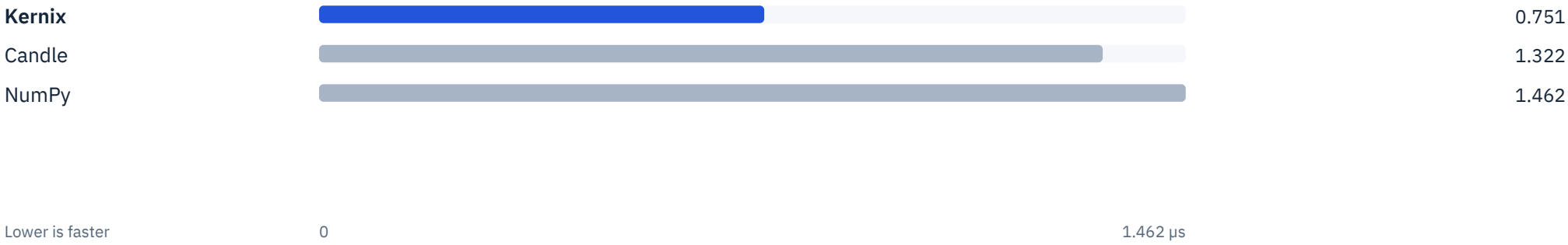
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.1 / Explicit parallel execution

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

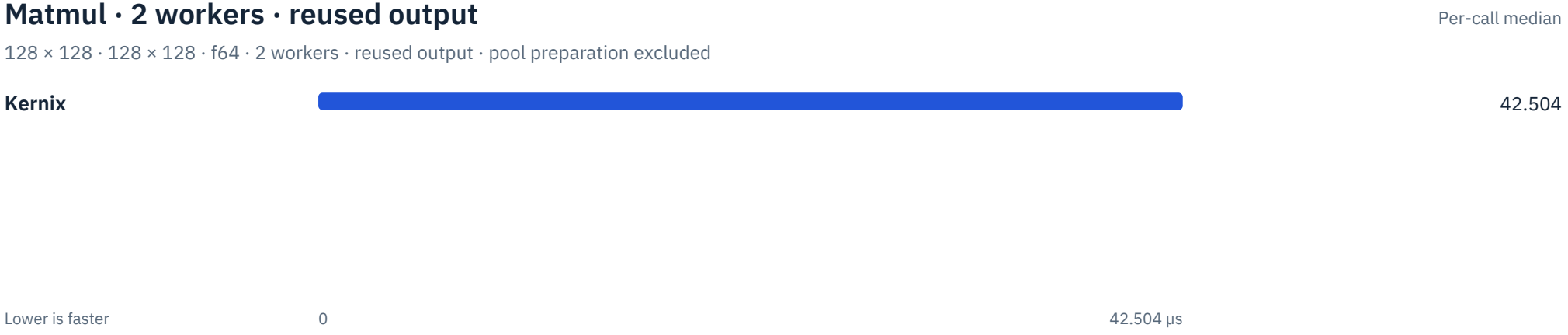
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Explicit parallel execution

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

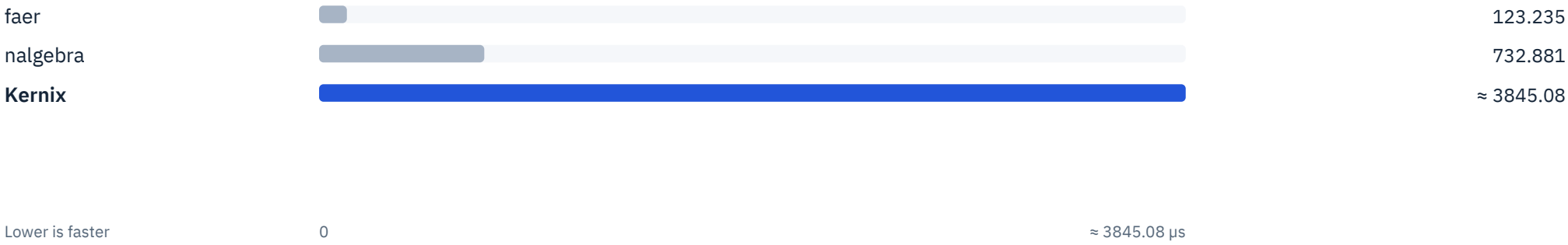
128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



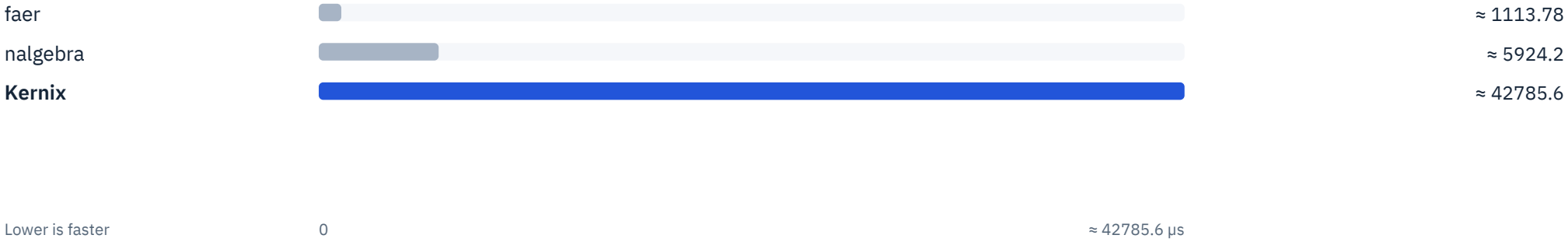
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

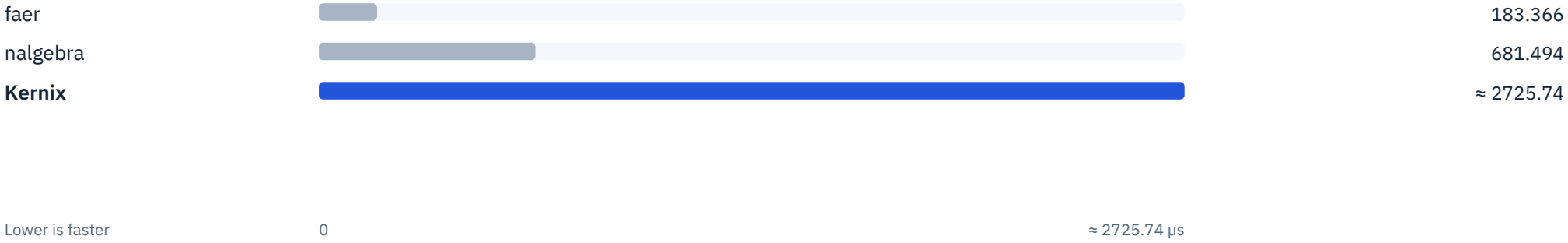
512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



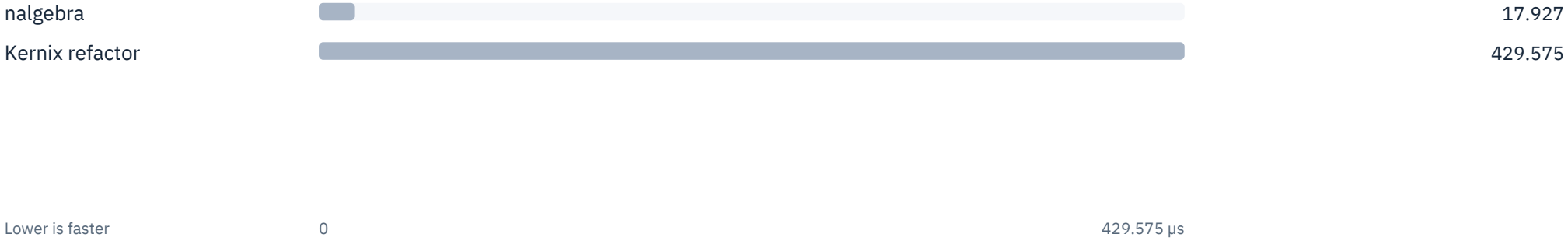
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

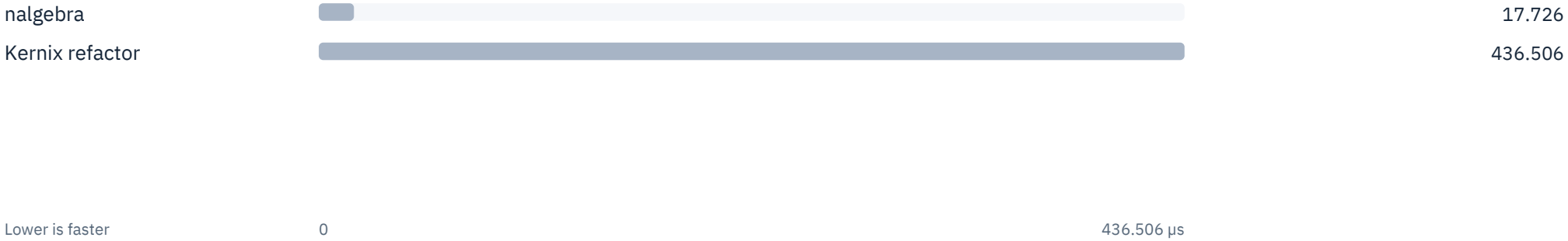
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

Per-call median



Thin QR

512 × 128 · f64 · 1 worker · allocated output

Per-call median



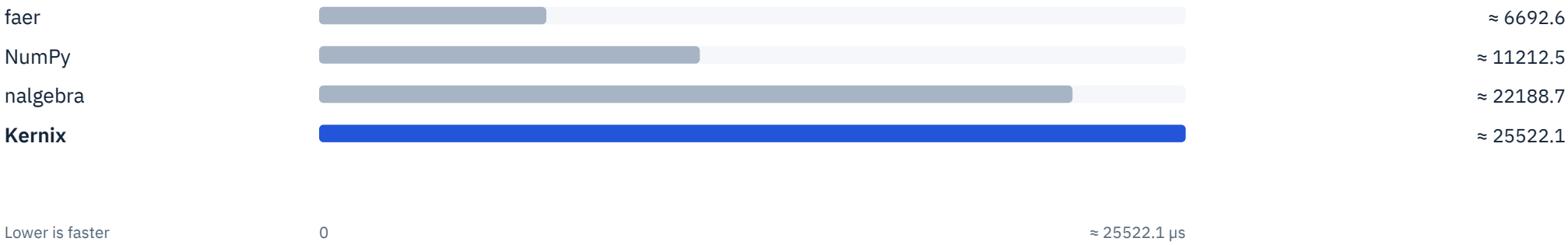
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

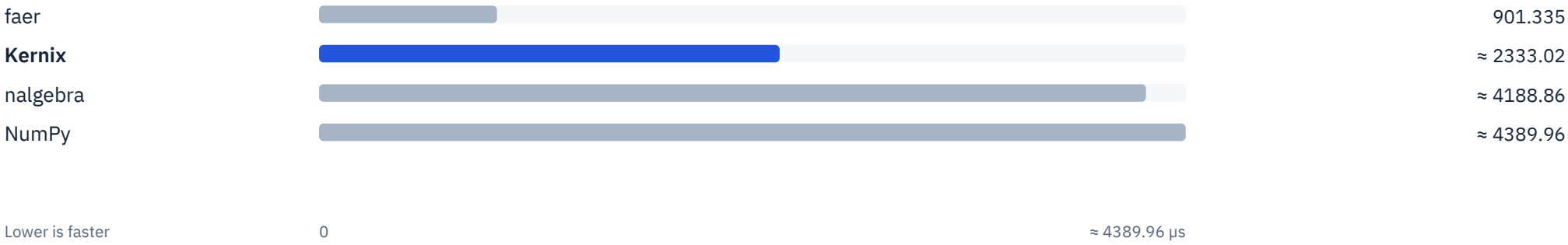
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



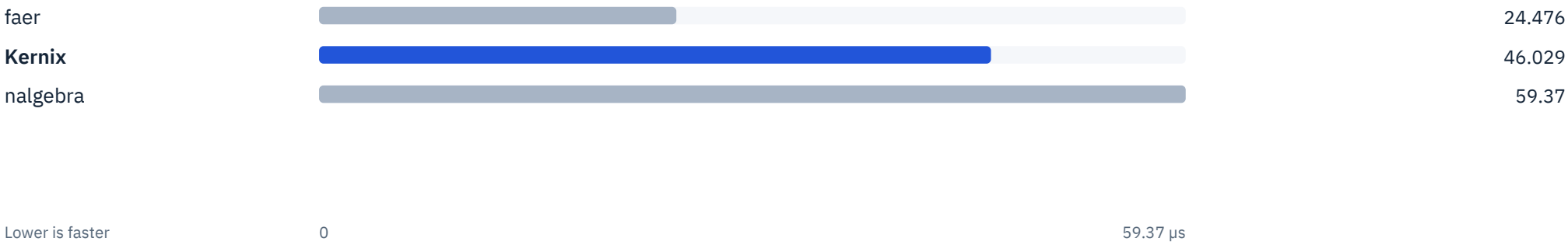
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



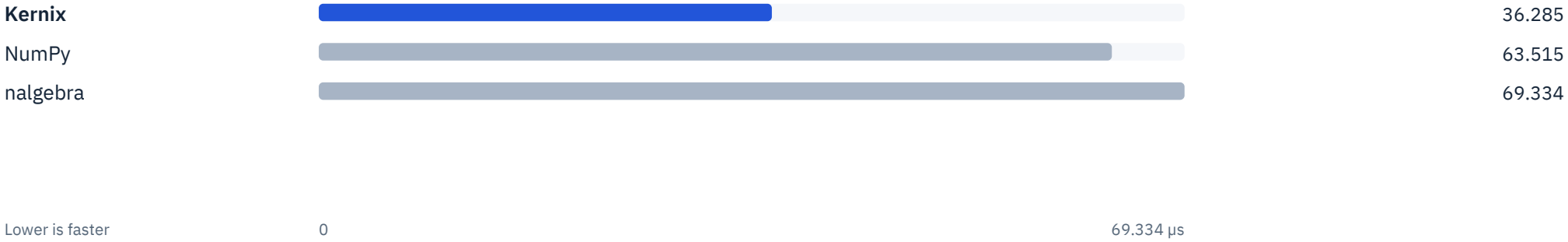
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

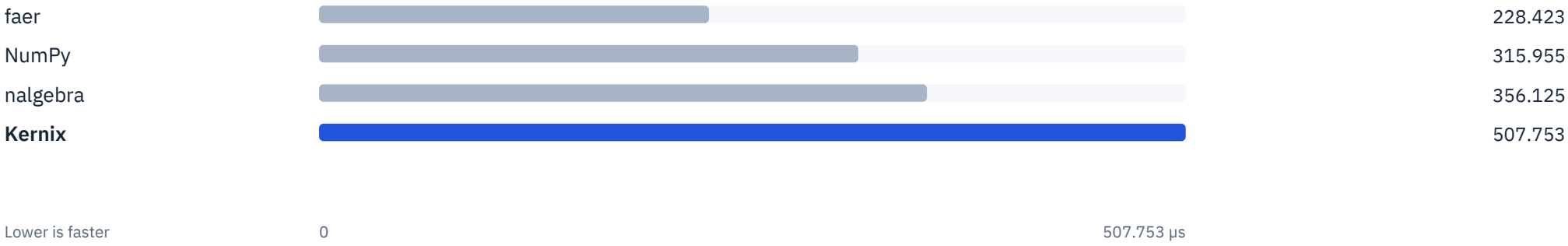
Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output

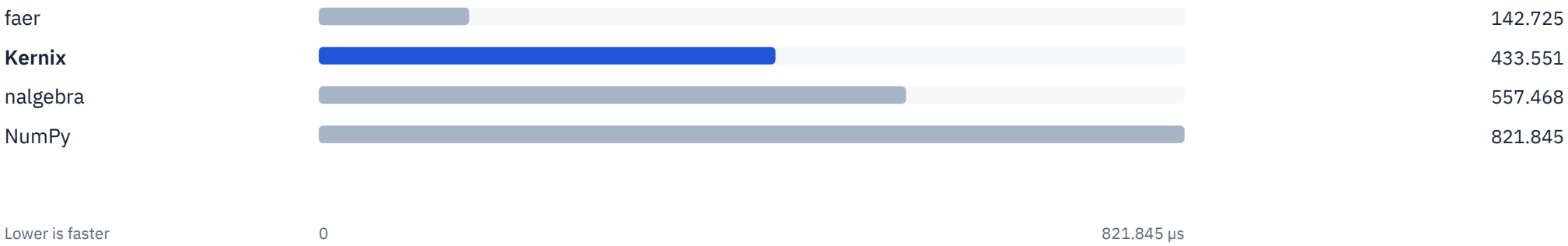


Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



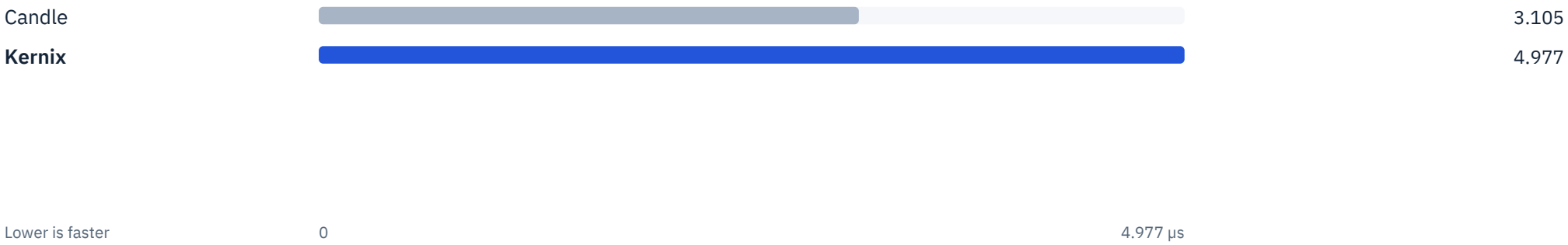
Kernix 0.11.1 / Training & autodiff

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Training & autodiff

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.1 / Training & autodiff

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

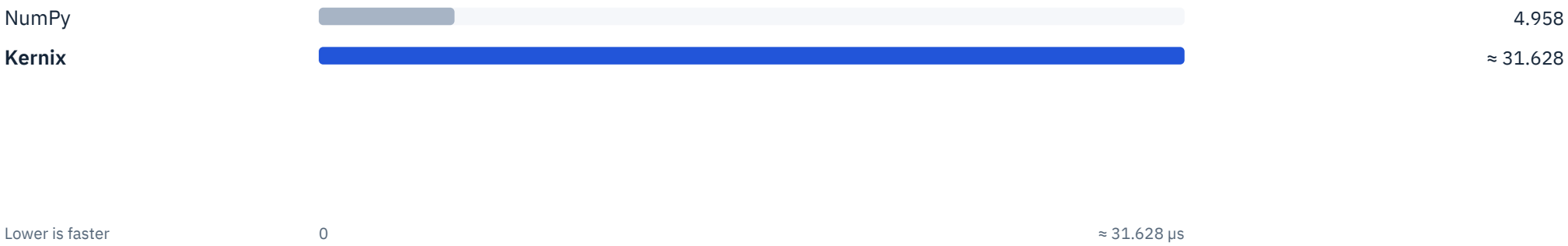


Kernix 0.11.1 / Tensor contractions

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



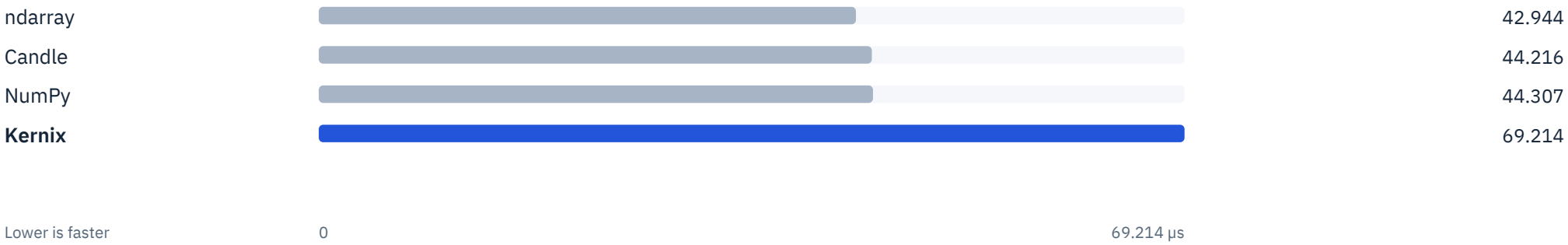
Kernix 0.11.1 / Statistics & activations

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Per-call median

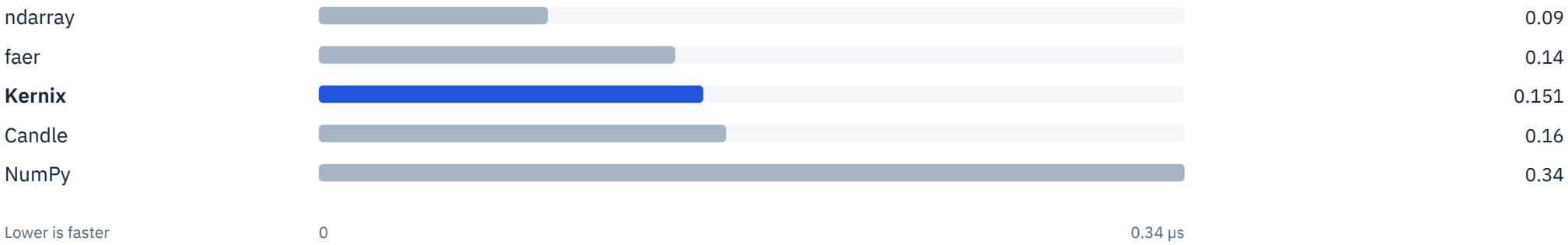


Kernix 0.11.1 / Tensor construction

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	49013	49.013
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	119730	119.73
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	21021	21.021
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	38367	38.367
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	69274	69.274
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	48102	48.102
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	33961	33.961
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	20781	20.781
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	990.9999999999999	0.9909999999999999
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1112	1.112

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	982.0000000000001	0.9820000000000001
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1392	1.392
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1412	1.412
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	772	0.772
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	4978	4.978
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	6350	6.35
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	9044	9.044
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	6370	6.37
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9004	9.004
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	4857	4.857

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	68873	68.873
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	48262	48.262
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	120280	120.28
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	68863	68.863
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	38478	38.478
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	121813	121.813
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	358018	358.018
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	330727	330.727
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	540022	540.022
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	544648	544.648

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	346461	346.461
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	338719	338.719
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	4447	4.447
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	3405	3.405
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	2975	2.975
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	2334	2.334
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	3435	3.435
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4537	4.537
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	21082	21.082
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	39760	39.76

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	40240	40.24
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	34232	34.232
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	34331	34.331
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	20030	20.03
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	329905	329.905
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	266700	266.7
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	150466	150.466
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	145118	145.118
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	263946	263.946
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	325860	325.86

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	1632	1.632
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1352	1.352
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	1922	1.922
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	1462	1.462
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	751	0.751
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	1322	1.322
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	10145	10.145
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	42504	42.504
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	287842	287.842
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	24687	24.687

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	3676	3.676
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	3606	3.606
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	9314	9.314
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	3845079	3845.079
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	732881	732.881
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	123235	123.235
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	42785584	42785.584
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	1113783	1113.783
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	5924202	5924.202
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	681494	681.494

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	2725745	2725.745
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	183366	183.366
Cholesky rank-one update / nalgebra 256 × 256 · f64	17927	17.927
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	429575	429.575
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	436506	436.506
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	17726	17.726
Cholesky rank-one update / nalgebra 512 × 512 · f64	95133	95.133
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	2325754	2325.754
Thin QR / nalgebra 512 × 128 · f64	2920398	2920.398
Thin QR / NumPy 512 × 128 · f64	2099615	2099.615

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 512 × 128 · f64	1210388	1210.388
Thin QR / Kernix 512 × 128 · f64	3512107	3512.107
Thin QR / Kernix 1024 × 256 · f64	25522079	25522.079
Thin QR / faer 1024 × 256 · f64	6692596	6692.596
Thin QR / NumPy 1024 × 256 · f64	11212453	11212.453
Thin QR / nalgebra 1024 × 256 · f64	22188670	22188.67
Cholesky factorization / nalgebra 512 × 512 · f64	4188855	4188.855
Cholesky factorization / faer 512 × 512 · f64	901335	901.335
Cholesky factorization / Kernix 512 × 512 · f64	2333015	2333.015
Cholesky factorization / NumPy 512 × 512 · f64	4389956	4389.956

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / nalgebra 96 × 32 · f64	59370	59.37
Column-pivoted QR / faer 96 × 32 · f64	24476	24.476
Column-pivoted QR / Kernix 96 × 32 · f64	46029	46.029
Column-pivoted QR / Kernix 256 × 64 · f64	531489	531.489
Column-pivoted QR / faer 256 × 64 · f64	216094	216.094
Column-pivoted QR / nalgebra 256 × 64 · f64	611399	611.399
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	36285	36.285
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	69334	69.334
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	63515	63.515
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	123435	123.435

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	118498	118.498
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	47852	47.852
Thin QR / faer 96 × 32 · f64	28984	28.984
Thin QR / NumPy 96 × 32 · f64	35564	35.564
Thin QR / nalgebra 96 × 32 · f64	34672	34.672
Thin QR / Kernix 96 × 32 · f64	39810	39.81
Thin QR / nalgebra 256 × 64 · f64	356125	356.125
Thin QR / Kernix 256 × 64 · f64	507753	507.753
Thin QR / faer 256 × 64 · f64	228423	228.423
Thin QR / NumPy 256 × 64 · f64	315955	315.955

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 64 × 64 · f64	10757	10.757
Cholesky factorization / Kernix 64 × 64 · f64	16104	16.104
Cholesky factorization / nalgebra 64 × 64 · f64	15133	15.133
Cholesky factorization / faer 64 × 64 · f64	7111	7.111
Cholesky factorization / NumPy 128 × 128 · f64	88803	88.803
Cholesky factorization / faer 128 × 128 · f64	27792	27.792
Cholesky factorization / Kernix 128 × 128 · f64	97277	97.277
Cholesky factorization / nalgebra 128 × 128 · f64	85699	85.699
Cholesky factorization / nalgebra 256 × 256 · f64	557468	557.468
Cholesky factorization / faer 256 × 256 · f64	142725	142.725

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 256 × 256 · f64	821845	821.845
Cholesky factorization / Kernix 256 × 256 · f64	433551	433.551
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	3105	3.105
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	4977	4.977
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	3826.0000000000005	3.8260000000000005
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2053	2.053
Model inference / Kernix batched normalized/sliced MLP · f64	1893	1.893
Model inference / Candle batched normalized/sliced MLP · f64	2955	2.955
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2423	2.423
Model recording + forward / Candle batched normalized/sliced MLP · f64	4226	4.226

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	3034	3.034
Model backward / Candle prepared graph → fresh gradients · f64	12739	12.739
Parameter update / Kernix transactional SGD, fixed gradients · f64	531	0.531
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	4958	4.958
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	31628.000000000004	31.628000000000004
Row softmax / ndarray 256 × 64 · f64	42944	42.944
Row softmax / Candle 256 × 64 · f64	44216	44.216
Row softmax / Kernix 256 × 64 · f64	69214	69.214
Row softmax / NumPy 256 × 64 · f64	44307	44.307
Create zeros / Kernix 64 × 64 · f32	151	0.151

Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	340	0.34
Create zeros / Candle 64 × 64 · f32	160	0.16
Create zeros / ndarray 64 × 64 · f32	90	0.09
Create zeros / faer 64 × 64 · f32	140	0.14
Create ones / faer 64 × 64 · f32	180	0.18
Create ones / NumPy 64 × 64 · f32	1242	1.242
Create ones / Kernix 64 × 64 · f32	230	0.23
Create ones / Candle 64 × 64 · f32	250	0.25
Create ones / ndarray 64 × 64 · f32	150	0.15